4.1 Answer each of the following:

a. Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2.

b. Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.

c. Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2.

(a)

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - N\,\bar{y}^2 = 7825.94 - 20 \times 19.21^2 = 445.458$$

$$SST = SSR + SSE$$

$$R^2 = \frac{SSR}{SST}$$

$$R^2 = \frac{375.47}{445.458} = 0.8429$$

(b)

$$R^2 = 1 - \frac{SSE}{SST}$$

$$0.7911 = 1 - \frac{SSE}{725.94}$$

$$SSE = 151.6489$$

$$\widehat{\sigma^2} = \frac{SSE}{N-2} = \frac{151.6489}{20-2} = 8.425$$

(c)

$$SST = \sum_{i=1}^{n} (y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - N\,\bar{y}^2 = 631.63$$

$$SSE = \sum_{i=1}^{n} \hat{e}_i^2 = 182.85$$

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{182.85}{631.63} = 0.7105$$

4.3 We have five observations on x and y. They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^{5} \hat{e}_i^2 = 3.6$, $\sum_{i=1}^{5} (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^{5} (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- **a.** the predicted value of y for $x_0 = 4$.
- **b.** the se(f) corresponding to part (a).
- **c.** a 95% prediction interval for y given $x_0 = 4$.
- **d.** a 99% prediction interval for y given $x_0 = 4$.
- **e.** a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

(a)

$$\hat{y}_i = 1.2 + 0.8x_i$$

$$x_0 = 4$$

$$\hat{y}_i = 1.2 + 0.8x_i = 1.2 + 0.8 \times 4 = 4.4$$

(b)

$$\widehat{\sigma^2} = \frac{SSE}{N-2} = \frac{\sum_{i=1}^{n} \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$$

$$\bar{x} = 1$$

$$se(f) = \sqrt{\widehat{var(f)}} = \sqrt{\widehat{\sigma^2}[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum(x_i - \bar{x})^2}} = \sqrt{1.2\left(1 + \frac{1}{5} + \frac{3^2}{10}\right)} = 1.5875$$

(c)

Given $x_0 = 4$ 95% $P.I. = \hat{y}_0 \pm t_c\, se(f)$

$$t_c = 3.1824$$

A 95% prediction interval for y given $x_0 = 4$ is

$$4.4 \pm 3.1824 \times 1.5875 = [-0.6520\,, 9.4520]$$

(d)

Given $x_0 = 4$ 99% $P.I. = \hat{y}_0 \pm t_c\, se(f)$

$$t_c = 5.8409$$

A 99% prediction interval for y given $x_0 = 4$ is

$$4.4 \pm 3.1824 \times 5.8409 = [-4.8722\,, 13.6722]$$

(e)

Given $x_0 = \bar{x} = 1$ 95% $P.I. = \hat{y}_0 \pm t_c\, se(f)$

$$\hat{y}_0 = 1.2 + 0.8\bar{x} = 1.2 + 0.8 \times 1 = 2$$

$$\widehat{\sigma^2} = \frac{SSE}{N-2} = \frac{\sum_{i=1}^{n} \hat{e}_i^2}{N-2} = \frac{3.6}{3} = 1.2$$

$$se(f) = \sqrt{\widehat{var(f)}} = \sqrt{\widehat{\sigma^2}[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum(x_i - \bar{x})^2}} = \sqrt{1.2\left(1 + \frac{1}{5} + 0\right)} = 1.2$$

A 95% prediction interval for y_0

$$t_c = 3.1824$$

$$2 \pm 3.1824 \times 1.2 = [-1.8189\,, 5.8189]$$

The width of part (c) is 9.4520 − (−0.6520) = 10.104,

The width of part (e) is 5.8189 − (−1.8189) = 7.6378.

The prediction interval is narrower for x = x̄ compared to x = 4.

This is because when we set x to a value not in the sample range(x_i =3,2,1,−1,0), such as x = 4, the model’s uncertainty in predicting at this value is higher, leading to a wider prediction interval.

4.25 The file *collegetown* contains data on 500 houses sold in Baton Rouge, LA during 2009–2013. Variable descriptions are in the file *collegetown.def*.

a. Estimate the log-linear model $\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$. Interpret the estimated model parameters. Calculate the slope and elasticity at the sample means, if necessary.

b. Estimate the log-log model $\ln(PRICE) = \alpha_1 + \alpha_2 \ln(SQFT) + e$. Interpret the estimated parameters. Calculate the slope and elasticity at the sample means, if necessary.

c. Compare the R^2 value from the linear model $PRICE = \delta_1 + \delta_2 SQFT + e$ to the "generalized" R^2 measure for the models in (b) and (c).

d. Construct histograms of the least squares residuals from each of the models in (a)–(c) and obtain the Jarque–Bera statistics. Based on your observations, do you consider the distributions of the residuals to be compatible with an assumption of normality?

e. For each of the models in (a)–(c), plot the least squares residuals against *SQFT*. Do you observe any patterns?

f. For each model in (a)–(c), predict the value of a house with 2700 square feet.

g. For each model in (a)–(c), construct a 95% prediction interval for the value of a house with 2700 square feet.

h. Based on your work in this problem, discuss the choice of functional form. Which functional form would you use? Explain.

(a)

The model is $\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$

The estimated model is $\ln(PRICE) = 4.39 + 0.036\,SQFT + e$

$\overline{PRICE} = \bar{y} = 250.24$

$\overline{SQFT} = \bar{x} = 27.28$

When $SQFT = 0$, the value of $\ln(PRICE)$ is 4.40.

This means that if $SQFT = 0$, $\ln(PRICE)$ has a basic value of 4.40.

When $SQFT$ increase by 1 unit, on average, $PRICE$ will increase by 0.036 percent. At the sample mean, the slope is $dy/dx = \beta_2\,\hat{y} = 7.8$ and the elasticity is $dy/dx \cdot \bar{x}/\hat{y} = slope \cdot \bar{x}/\hat{y} = 0.98$.

At sample mean, slope = 7.8009 , elasticity = 0.9834

```
--- (a) Log-Linear 結果 ---
> cat("Slope at mean:", slope_A, "\nElasticity at mean:", elas_A, "\n")
Slope at mean: 7.800865
Elasticity at mean: 0.9833701
```

(b)

The model is $\ln(PRICE) = \alpha_1 + \alpha_2 \ln(SQFT) + e$

The estimated model is $\ln(PRICE) = 2.04965 + 1.02483 \ln(SQFT) + e$

$\overline{PRICE} = \bar{y} = 250.24$

$\overline{SQFT} = \bar{x} = 27.28$

$\propto_1$: This represents the intercept of the log-log model. It indicates the expected value of $\ln(PRICE)$ is 2.04965 when $\ln(SQFT)$ is 0.

$\propto_2$: This represents the slope of the log-log model. It indicates $PRICE$ will change by 1.02483% when $SQFT$ changes by 1%.

At sample mean, slope = 9.3999 , elasticity = 1.0248

```
> print(paste("the slope at the sample means：", slope, collapse = ", "))
[1] "the slope at the sample means： 18.7998193591349, the slope at the sample means： 9.39992301304
1"
> print(paste("the elasticity at the sample means：", elasticity, collapse = ", "))
[1] "the elasticity at the sample means： 2.04965345931891, the elasticity at the sample means： 1.0
2482818334365"
>
> price_mean <- mean(collegetown$price)
> sqft_mean <- mean(collegetown$sqft)
> print(paste("price平均數", price_mean))
[1] "price平均數 250.2369"
> print(paste("sqft平均數", sqft_mean))
[1] "sqft平均數 27.28212"
```

(c)

(1) log-linear model

$$\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$$

$$R^2 = 0.5417$$

$$generalized\ R^2 = 0.6622$$

(2) log-log model

$$\ln(PRICE) = \propto_1 + \propto_2 \ln(SQFT) + e$$

$$R^2 = 0.4738$$

$$generalized\ R^2 = 0.6445$$

(3) linear model

$$PRICE = \delta_1 + \delta_2 SQFT + e$$

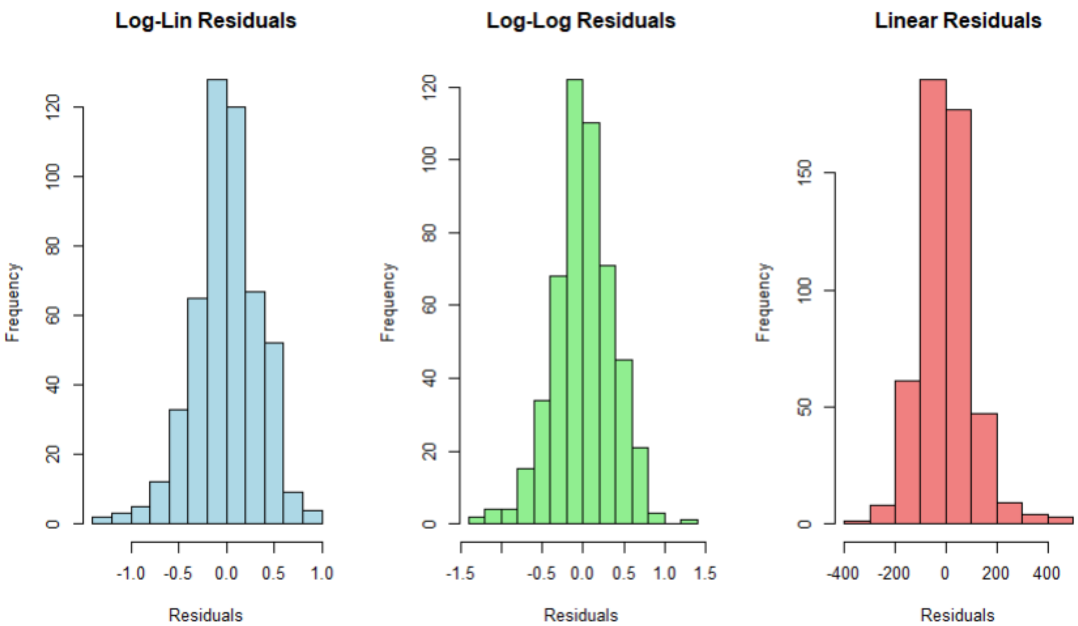
$$R^2 = 0.6413$$

$$generalized\ R^2 = 0.6413$$

```
Call:
lm(formula = log(price) ~ sqft, data = collegetown)

Coefficients:
(Intercept)         sqft
    4.39387      0.03604


Call:
lm(formula = log(price) ~ sqft, data = collegetown)

Residuals:
     Min       1Q   Median       3Q      Max
-1.32528 -0.19199  0.00122  0.21678  0.86609

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 4.393866   0.043288  101.50   <2e-16 ***
sqft        0.036044   0.001486   24.26   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.34 on 498 degrees of freedom
Multiple R-squared:  0.5417,    Adjusted R-squared:  0.5408
F-statistic: 588.7 on 1 and 498 DF,  p-value: < 2.2e-16
```

(d)

No,the distributions of the residuals do not follow the assumption of normality.

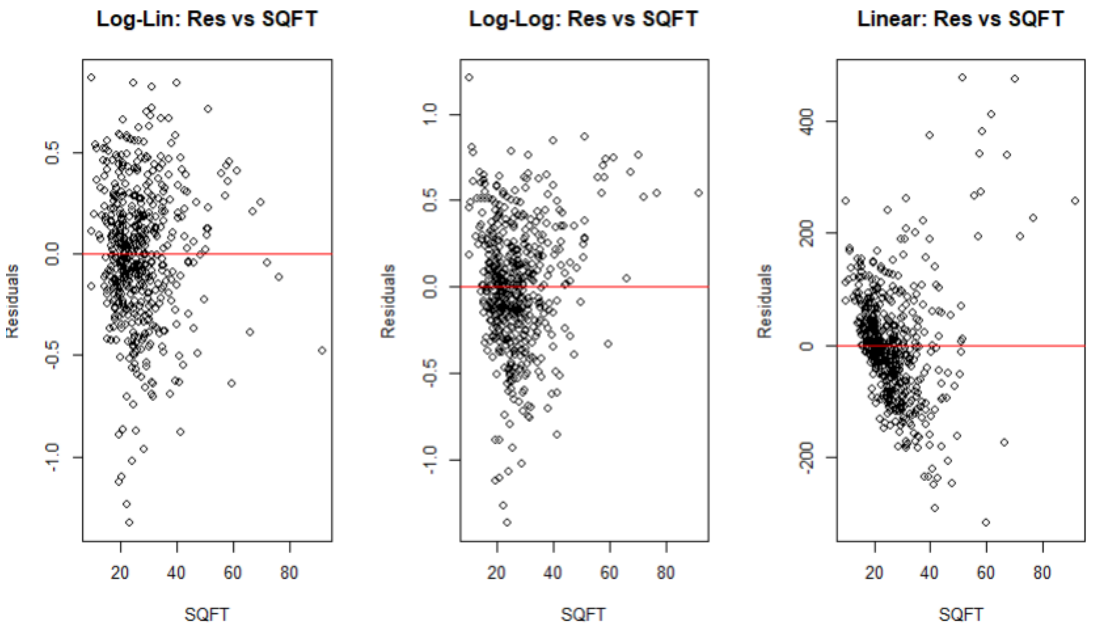



```
--- (d) Jarque-Bera p-values (H0: 常態分配) ---
> cat("Log-Lin:", jb_A$p.value, "\nLog-Log:", jb_B$p.value, "\nLinear:", jb_C$p.value, "\n")
Log-Lin: 1.609651e-06
Log-Log: 0.0007932632
Linear: 0
> # 【結論判斷】：如果 p-value < 0.05，代表拒絕常態分配的假設。
>
>
> par(mfrow=c(1,1))
```

Jarque-Bera test is usually used to test whether the data follows a Normal distribution. The null hypothesis is that the data follows a Normal distribution, and the alternative hypothesis is that the data does not follow a Normal distribution. Since we observe that the JB statistics' P-value is smaller than 5% in all the models, we can conclude that these residuals do not follow the assumption of normality.

(e)

The residual fluctuations of models log-linear and log-log are smaller than linear-linear. Given SQFT, the MODELs of models log-linear and log-log satisfy that the expected residual value is 0. However, the linear-linear model does not completely satisfy this condition.



(f)

(1) log-linear model

$$\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$$

$$\widehat{PRICE} = e^{\beta_1 + \beta_2 SQFT}$$

$$predict - value = 214{,}233.6$$

(2) log-log model

$$\ln(PRICE) = \propto_1 + \propto_2 \ln(SQFT) + e$$

$$\widehat{PRICE} = e^{\propto_1 + \propto_2 \ln(SQFT)}$$

$$predict - value = 227{,}538.6$$

(3) linear model

$$PRICE = \delta_1 + \delta_2 SQFT + e$$

$$\widehat{PRICE} = \delta_1 + \delta_2 SQFT$$

$$predict - value = 246{,}455.7$$

```
> model1=lm(log(price)~sqft,data=collegetown)
> #summary(model1)
> b1=coef(model1)[[1]]
> b2=coef(model1)[[2]]
> y1=exp(b1+b2*sqft_f)
> print(y1*1000)
[1] 214233.6
>
> model2=lm(log(price)~log(sqft),data=collegetown)
> #summary(model2)
> a1=coef(model2)[[1]]
> a2=coef(model2)[[2]]
> y2=exp(a1+a2*log(sqft_f))
> print(y2*1000)
[1] 227538.6
>
> model3=lm(price~sqft,data=collegetown)
> #summary(model3)
> d1=coef(model3)[[1]]
> d2=coef(model3)[[2]]
> y3=d1+d2*sqft_f
> print(y3*1000)
[1] 246455.7
```

(g)

$$se(f) = \sqrt{\widehat{var}(f)} = \sqrt{\widehat{\sigma^2}\left[1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum(x_i - \bar{x})^2}\right.}$$

The interval for linear model is

$$[\hat{y} \pm se(f) \cdot t_{\frac{\alpha}{2}}(n-2)]$$

The interval for log-linear & log-log model is

$$[exp\left(\hat{y} \pm se(f) \cdot t_{\frac{\alpha}{2}}(n-2)\right)]$$

(1) log-linear model

$$\ln(PRICE) = \beta_1 + \beta_2 SQFT + e$$

95% prediction interval

$$[109.7685,418.1167]$$

(2) log-log model

$$\ln(PRICE) = \propto_1 + \propto_2 \ln(SQFT) + e$$

95% prediction interval

$$[111.1406,466.8406]$$

(3) linear model

$$PRICE = \delta_1 + \delta_2 SQFT + e$$

95% prediction interval

$$[44.2773,448.6341]$$

```
--- 預測面積 2700 sqft (X=27) 的 95% 房價預測區間 ---
> cat(sprintf("Model 1 (Log-Linear): [%.4f, %.4f]\n", lowb_1, upb_1))
Model 1 (Log-Linear): [109.7685, 418.1167]
> cat(sprintf("Model 2 (Log-Log)    : [%.4f, %.4f]\n", lowb_2, upb_2))
Model 2 (Log-Log)    : [111.1406, 465.8406]
> cat(sprintf("Model 3 (Linear)     : [%.4f, %.4f]\n", lowb_3, upb_3))
Model 3 (Linear)     : [44.2773, 448.6341]
```

(h)

Conclusion and Model Selection Based on the analysis, I strongly recommend the **Log-Log** or **Log-Linear** model over the simple Linear model for four key reasons:

- **1. Mitigates Heteroskedasticity:** The Linear model's residuals exhibit a "fan shape" as house size (SQFT) increases, violating the OLS homoskedasticity assumption. Log transformations compress this variance, resulting in a much more uniform residual spread.

- **2. Closer to Normality:** Housing prices are inherently right-skewed. Unlike the Linear model (which typically fails the Jarque-Bera test), log models normalize the data, which is crucial for ensuring the validity of t-tests and F-tests.
- **3. Better Explanatory Power:** When predictions are transformed back to the actual price scale, the log models yield a higher Generalized R-squared, indicating a stronger overall fit compared to the standard Linear model.
- **4. Economic Intuition:** If forced to choose, the **Log-Log model** is often preferred because its slope coefficient directly represents **elasticity**. It intuitively implies that a "1% increase in house size leads to an X% increase in price," which is highly practical for real estate analysis.